

Theoretical Astrophysics I: Stellar Structure and Evolution

Lecture 7: Simple models of stellar structure

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Recap

- ▶ Now, we understand the main processes that govern stellar structure (*not yet evolution!*).
- ▶ To recap our main assumptions:
- ▶ Stars are in hydrostatic equilibrium.
- ▶ Energy is transported either by radiation or convection
- ▶ Hopefully we understand equation of state (connection between T, ρ, P)
- ▶ Hopefully we can calculate the opacity from there
- ▶ And hopefully we can calculate the rate of energy production.

Equations of stellar structure

Here they are, written in terms of r as independent variable:

$$\frac{dP}{dr} = -\rho \frac{Gm}{r^2} \quad (1)$$

$$\frac{dm}{dr} = 4\pi r^2 \rho \quad (2)$$

$$\frac{dT}{dr} = -\frac{3}{4ac} \frac{\kappa \rho}{T^3} \frac{F}{4\pi r^2} \quad (3)$$

$$\frac{dF}{dr} = 4\pi r^2 \rho q \quad (4)$$

$$P = \frac{\mathcal{R}}{\mu_I} \rho T + P_e + \frac{1}{3} a T^4 \quad (5)$$

$$\kappa = \kappa_0 \rho^a T^b \quad (6)$$

$$q = q_0 \rho^m T^n \quad (7)$$

Equations of stellar structure

Count the unknowns, count the equations, is the system, in principle, solvable?

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Equations of stellar structure

Four of these equations are differential equations. Three are algebraic. Differential equations are of the first order, so each one needs one boundary condition. These are, generally speaking.

- ▶ At $r = 0$ (center), $m = 0$, $F = 0$.
- ▶ At $m = M$, $P = 0$ and $F = L$.

It turns out that two physical parameters that govern the structure a so-called ZAMS (zero age main sequence star) star are the **total mass** and **chemical composition**. More about that in the next class.

Can someone tell me why we could not say at $r = R$, $m = M$ and $P = 0$ and be done with it?

Equations of stellar structure

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Can someone tell me why we could not say at $r = R$, $m = M$ and $P = 0$ and be done with it? Because we do not know what R is. When solving the equations of stellar structure, we don't know in advance how big the star will be!

General assumptions

To ease our job, some simple things can, generally be assumed

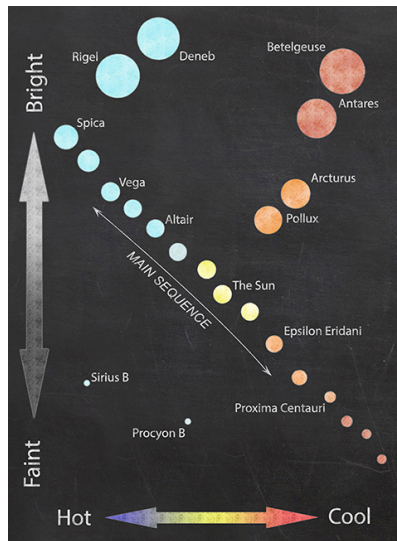
- ▶ Constant μ , either for ZAMS or completely evolved stars.
- ▶ When $q =$, F is constant.
- ▶ P and ρ and T can all be safely assumed to monotonously decrease with r (m).
- ▶ It is super helpful if we can find combinations of quantities that stay approximately constant over the stellar radius.

Now - be mindful that the situation can be even more dire - as the opacity and nuclear rate production can be very complicated and no analytical solution is possible.

In modeling, in general, we solve everything by numerics.

Our end goal

- ▶ Ideally, if we manage to calculate the stellar model, and repeat the calculation for the models of different mass and luminosity, we should get something like HR diagram.
- ▶ It should also give us some very basic insights into the stellar evolution, even if we do not numerically *evolve* the star.
- ▶ Remember once again, that equations of stellar structure, in a way, tell us what kinds of stars are permitted to exist by physical laws!



Credits: the Open University

Further simplifications

These equations were written more than 100 years ago. Back then we could not calculate anything numerically (well, almost anything). Now, let's look at the differential equations again:

$$\frac{dP}{dr} = -\rho \frac{Gm}{r^2}$$

$$\frac{dm}{dr} = 4\pi r^2 \rho$$

$$\frac{dT}{dr} = -\frac{3}{4ac} \frac{\kappa \rho}{T^3} \frac{F}{4\pi r^2}$$

$$\frac{dF}{dr} = 4\pi r^2 \rho q$$

(8)

First two equations only involve m, P, ρ . What would I need to close the system?

Further simplifications

What would I need to close the system?

Equation of state! But such an equation of state, where T does not appear, that is where P depends only on ρ (μ can also appear as it is a given constant, but discuss this for a sec).

$$\begin{aligned}\frac{dP}{dr} &= -\rho \frac{Gm}{r^2} \\ \frac{dm}{dr} &= 4\pi r^2 \rho \\ P &= K\rho^\gamma\end{aligned}\tag{9}$$

Where we typically define n as $\gamma = 1 + 1/n$.

So, now we have closed the system of equations, and we can go to town! :-)

The ensuing models are called **polytropes**, and can be solved analytically only in a handful of cases (for specific n 's) - as you have seen in your homework.

Sketching the solution

$$\begin{aligned}\frac{dP}{dr} &= -\rho \frac{Gm}{r^2} \\ \frac{dm}{dr} &= 4\pi r^2 \rho\end{aligned}\tag{10}$$

Multiply first equation by $\frac{r^2}{\rho}$, differentiate with respect to dr and substitute into the equation two, you will get:

$$\frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho} \frac{dP}{dr} \right) = -4\pi G \rho\tag{11}$$

And good old: $P = K\rho^\gamma$

Sketching the solution

Now, ρ is the only unknown!

$$\frac{(n+1)K}{4\pi Gn} \frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho^{(n-1)/n}} \frac{d\rho}{dr} \right) = -\rho \quad (12)$$

We have a second order differential equation, with only one unknown (need two boundary conditions)!

This is where the problem lies, discuss!

Sketching the solution

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This is a two-point boundary problem, these often arise in stellar physics. Namely, we have two different boundary conditions at two different points.

For example: $(\frac{d\rho}{dr})_{r=0} = 0$ and $(\rho)_{r=R} = 0$.

This gives rise to additional complications. But today we will not actually work on the solution.

Sketching the solution

$$\frac{(n+1)K}{4\pi Gn} \frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho^{(n-1)/n}} \frac{d\rho}{dr} \right) = -\rho$$

We have a second order differential equation, with only one unknown (need two boundary conditions)!

The complete solution is determined by K, n, R (R is not ideal, but we can “tune” it so that R reproduces desired mass). Once we solve the problem we can calculate everything from ρ . Even the equations that we did not start from (e.g. Flux).

Sketching the solution

Additional steps in the solution are, we define θ , which serves as a scaling factor:

$$\rho = \rho_c \theta^n \quad (13)$$

and

$$\left[\frac{(n+1)K}{4\pi G \rho_c^{(n-1)/n}} \right] = \alpha^2 \quad (14)$$

which defines:

$$r = \alpha \xi \quad (15)$$

And leads to the **Lane - Emden equation of index n** :

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n \quad (16)$$

With boundaries: $\theta = 1, \frac{d\theta}{d\xi} = 0$ at $\xi = 0$.

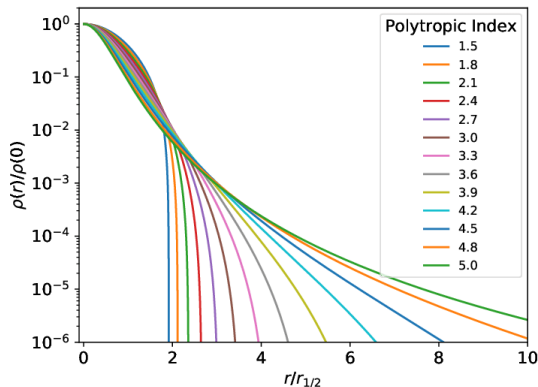
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Now, we can integrate numerically from the center outwards. Solutions with $n < 5$ decrease monotonically and have zero at a finite ξ (not guaranteed), which then defines the stellar radius.



Credits: Sanchez Almeida et al 2020

Mass-Radius relationship

After solving for given n , we can integrate what we need to find the total mass radius, etc.
Important ensuing relationship is:

$$\frac{GM^{n-1}}{M_n} \frac{R^{3-n}}{R_n} = \frac{[(n+1)K]^n}{4\pi G} \quad (18)$$

Where R_n and M_n are constants for given n (see table 5.1 in the book).
Two special cases $n = 3$ and $n = 1$.

Why? Discuss!

Mass-Radius relationship

$$\frac{GM^{n-1}}{M_n} \frac{R^{3-n}}{R_n} = \frac{[(n+1)K]^n}{4\pi G} \quad (19)$$

Two special cases $n = 3$ and $n = 1$.

For $n = 3$ mass is independent of radius of the star and only depends on K .

For $n = 1$ radius is independent of mass of the star and only depends on K .

But in between, we have the following:

$$R^{3-n} \propto \frac{1}{M^{n-1}} \quad (20)$$

Both exponents are positive, so more massive stars are smaller!

They also then have higher central pressures, burn faster, and eventually live shorter (need some imagination, but it's true).

Pressure - density relationship in the center

For completeness (look at the book for details).

If we now express K through M and R , and “compress” some constants, we can get:

$$P_c = \text{const} B_n M^{2/3} \rho_c^{4/3} \quad (21)$$

B_n itself turns out to vary slowly with n , so the relationship is almost universal. We will use it soon to discuss the stellar evolution.

Degenerate gas

Polytropes with $n = 1.5$ do exist. These are the stars supported by degenerate pressure of electrons. So called white dwarves. They are extremely dense: $10^5 \times$ solar density.

For $n = 1.5$, we get:

$$R \propto M^{-1/3} \quad (22)$$

implying:

$$\bar{\rho} \propto M^2 \quad (23)$$

Meaning that some average density of the star increases quadratically with mass.

This compression eventually causes the gas to become *relativistic*, and the polytrope to shift to $n = 3$.

Relativistic degenerate gas

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But for $n = 3$ we have had, according to:

$$\frac{GM^{n-1}}{M_n} \frac{R^{3-n}}{R_n} = \frac{[(n+1)K]^n}{4\pi G} \quad (24)$$

M has only one value, independent of R .

The interpretation for this is tricky.

We can say that there is an upper limit for the mass at which the gas can reach $n = 3$.

Smaller masses do not reach exactly $n = 3$, but higher masses are *impossible*.

Are they really impossible?

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But for $n = 3$ we have had, according to:

$$\frac{GM^{n-1}}{M_n} \frac{R^{3-n}}{R_n} = \frac{[(n+1)K]^n}{4\pi G} \quad (25)$$

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Smaller masses do not reach exactly $n = 3$, but higher masses are *impossible*.

Not quite - they are impossible for the assumed equation of state. More dense structures exist, these are named neutron stars.

Chandrasekhar

Subrahmanyan Chandrasekhar (1910 - 1985), was Indian-American astrophysicist, one of the fathers of this theory and a nobel laureate (shared with Fowler).

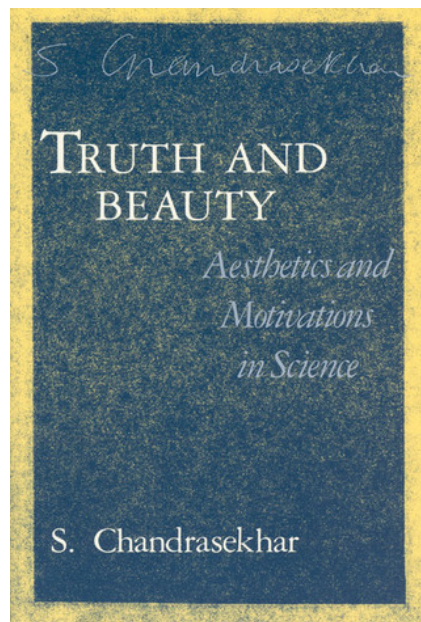
He did the derivation above (more or less), and finally concluded that the limiting mass is, in physical units:

$$M_{\text{Ch}} = 5.83\mu_e^{-2}M_{\odot} \quad (26)$$

Now for a star consisting of He, C, O, purely ionized $\mu_e = 2$, and limiting mass is 1.46 solar masses.

For iron, $\mu_e = 2.15$, and the mass is 1.26 solar masses.

Discuss a bit the values of μ and these masses. How can we verify Chandrasekhar's finding?



Including radiative energy transport

In the starting days of stellar structure theory, it was commonly assumed that transport is purely radiative. It was hard to model convection, even with mixing-length theory. We will now include radiative transport in our modeling:

$$\frac{dP}{dr} = -\rho \frac{Gm}{r^2}$$

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leads to:

$$\frac{dP_{\text{rad}}}{dP} = \frac{\kappa F}{4\pi c G m}\tag{28}$$

As radiation pressure cannot be larger than the total pressure - it means there is a limit to the radiative flux of the star!

$$\frac{dP_{\text{rad}}}{dP} = \frac{\kappa F}{4\pi c G m} \quad (29)$$

As radiation pressure cannot be larger than the total pressure - it means there is a limit to the radiative flux of the star, that is:

$$kF < 4\pi cGm \quad (30)$$

Or at the surface:

$$L < \frac{4\pi cGM}{\kappa} \quad (31)$$

which leads to definition of so called Eddington luminosity:

$$L_{\text{Edd}} = 3.2 \times 10^4 \frac{M}{M_{\odot}} \left(\frac{\kappa_{es}}{\kappa} \right) L_{\odot} \quad (32)$$

Important: As there is another connection between L and M , this puts a limit on the maximum mass of the star. It is indeed the case. Very massive stars can exist, but live shortly and lose **a lot** of mass through stellar wind.

Standard model

Standard (or Eddington) model, assumes the energy transport via radiation, and identifies a quantity that is assumed to vary slowly with radius. First we define a function $\eta(r)$ as:

$$\frac{F(r)}{m(r)} = \eta(r) \frac{L}{M} \quad (33)$$

Then we recall the relationship between radiation pressure and total pressure and cast it using η :

$$\frac{dP_{\text{rad}}}{dP} = \frac{L}{4\pi cGM} \kappa \eta \quad (34)$$

Then we make a bold argument! $\eta = 1$ at the surface and increases inward. κ is largest at the surface (recall plots of opacity from the last class) and decreases inwards. **Eddington assumed** $\kappa\eta = \text{const.}$

$$\frac{dP_{\text{rad}}}{dP} = \frac{L}{4\pi cGM} \kappa \eta \quad (35)$$

Eddington assumed $\kappa \eta = \text{const.}$ Integration of the above equation from the center to the surface leads to:

$$P_{\text{rad}} = \frac{\kappa_s L}{4\pi cGM} P. \quad (36)$$

Let's recall the definition of the parameter β : $P_{\text{rad}} = (1 - \beta)P$.

Here $\beta = 0$ for radiation dominated pressure and $\beta = 1$ for gas dominated pressure. Now, assumption of a constant β allows us to cast everything in terms of radiation pressure *even though the star is not dominated by the radiation pressure!*.

Standard model

Now we assume ideal gas and get:

$$\frac{\mathcal{R}}{\beta\mu}\rho T = \frac{aT^4}{3(1-\beta)} \quad (37)$$

which gives us relationship between T and ρ :

$$T = \left[\frac{3\mathcal{R}(1-\beta)}{a\mu\beta} \right] \rho^{1/3} \quad (38)$$

which immediately leads to:

$$P = K\rho^{4/3} \quad (39)$$

where K depends on physical constants, μ and β . Here we go, another polytrope with $n = 3!!!$

Recall that in this case the mass is constant and uniquely determined by K :

$$M = 4\pi M_3 \left(\frac{K}{\pi G} \right)^{3/2} \quad (40)$$

Which means the mass depends on β , which in turn depends on luminosity of the star relative to Eddington luminosity. We have reached somehow the **mass - luminosity relationship!**
But before, we will reach the famous *Eddington quartic equation*:

$$1 - \beta = 0.003 \left(\frac{M}{M_\odot} \right)^2 \mu^2 \beta^4. \quad (41)$$

It turns out that the stars exist in the regime where β is somewhere between 0 and 1. [Let's read Eddington's statement from the book.](#)

Mass-Luminosity Relationship

Finally, substituting the $1 - \beta$ we just found into:

$$L = \frac{4\pi cGM}{\kappa_s}(1 - \beta) \quad (42)$$

we get the classic **mass - luminosity relationship**:

$$\frac{L}{L_\odot} = \frac{4\pi cGM_\odot}{\kappa_s L_\odot} 0.003 \mu^4 \beta (M, \mu)^4 \left(\frac{M}{M_\odot}\right)^3 \quad (43)$$

which shows that the luminosity scales as mass to the power of three, plus some scaling due to the β .

Mass-Luminosity Relationship

